

DISORDERS OF THE ADRENAL GLAND

Cushing Syndrome

Excessive production of endogenous cortisol manifests clinically as **Cushing syndrome**. Similar clinical findings can also result from chronic exogenous administration of glucocorticoids. In normal physiology, the hypothalamic-pituitary-adrenal (HPA) axis regulates cortisol production: corticotropin-releasing hormone (CRH) from the hypothalamus stimulates the anterior pituitary to produce corticotropin (ACTH). ACTH is derived from a larger precursor molecule called proopiomelanocortin (POMC). POMC is cleaved into several biologically active peptides, including melanocyte-stimulating hormone (MSH) and ACTH. ACTH then stimulates the adrenal cortex to produce cortisol. Cortisol exerts negative feedback on both the hypothalamus and pituitary, suppressing further CRH and ACTH release.

EPIDEMIOLOGY, ETIOLOGY, AND PATHOGENESIS

The overall incidence of Cushing syndrome is approximately 10 cases per million people per year, with a higher prevalence in women.

Cushing syndrome is categorized into two main types based on etiology:

1. **ACTH-dependent Cushing syndrome:**
2. Most commonly caused by a pituitary microadenoma secreting excessive ACTH, known as **Cushing disease**.
3. Less commonly, **ectopic ACTH production** occurs, typically associated with neuroendocrine tumors such as pulmonary carcinoid tumors, though other malignancies may also be responsible.
4. **ACTH-independent Cushing syndrome:**

5. Results from autonomous overproduction of cortisol by the adrenal cortex, independent of pituitary regulation.
6. Causes include:
 - Adrenal adenoma (benign or malignant)
 - Bilateral adrenal hyperplasia
 - Primary pigmented nodular adrenocortical disease (as part of **Carney complex**)
 - McCune-Albright syndrome

CLINICAL FINDINGS: CUTANEOUS LESIONS

The clinical presentation of Cushing syndrome can be insidious. Cutaneous manifestations are common and may provide early diagnostic clues when correlated with systemic symptoms.

Key skin findings include:

- **Easy bruisability** and **skin atrophy** due to collagen loss
- **Thin, fragile skin** with a translucent appearance
- **Mild hypertrichosis**, often of lanugo-type hair
- **Acne**, which may be moderate to severe
- **Hirsutism** and **severe acne** suggest possible coexisting androgen excess, raising suspicion for an androgen-secreting adrenal tumor
- **Violaceous striae**: wide (>1 cm), purple or reddish-purple striae on the abdomen, thighs, or proximal extremities — highly suggestive of Cushing syndrome (Fig. 152-15)

Careful evaluation of these dermatologic signs in the context of systemic features (e.g., central obesity, hypertension, glucose intolerance, moon facies, proximal muscle weakness) can facilitate early diagnosis.